# Position, distance, displacement

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Distinguish a location, a total path length, and a change in position.

**SUCCESS CRITERIA - I can...**

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

**Core:** teacher selects from 1-9. **Stretch:** 10 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Positions are **15**, **10**, **5 m** at **0**, **2**, **4 s**.
1. **Predict position** at **7 s**.
2. **Give the rate**, something you are treating as true and a future test.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $\Delta x = x_f - x_i$; distance = sum of path lengths. **Choose an origin** and positive direction. i = initial; f = final.

**Worked reasoning:** A walker starts at **2 m**, reaches **6 m**, then ends at **4 m**. Distance=**4**+**2**=**6 m**. Displacement=**4**-**2**=**2 m** right.

**Your turn:** The walker goes from the **2 m** floor mark to the **6 m** mark, then back to **4 m**. On the number line, draw the two legs above the axis and a separate start-to-finish arrow below it. **Label each length**.

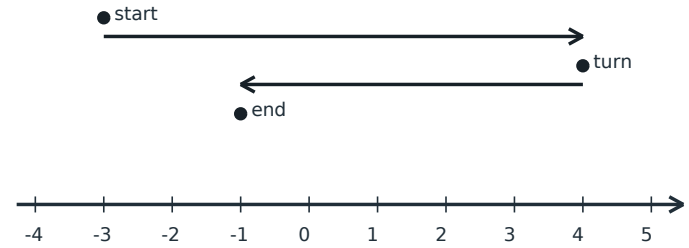



Position (m); right is positive

**Marzano check - circle:** ① ② ③

## 03 Track a signed route

A robot moves along an *east*-*west* test track. *East* is positive. It starts at **-3 m**, reaches **+4 m**, then stops at **-1 m**. **Use the route** to **find total distance** and displacement.




Position (m); right is positive

**Marzano check - circle:** ① ② ③

## 04 Change the coordinate convention

For the route in **3**, reverse the positive direction while keeping the origin fixed.
1. **Give the new endpoint coordinates** and displacement.
2. What stays unchanged?

**Marzano check - circle:** ① ② ③

# Position, distance, displacement

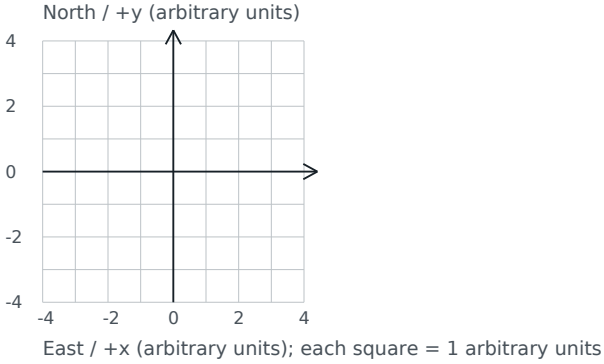
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Continue down each column.

### 05 A closed path

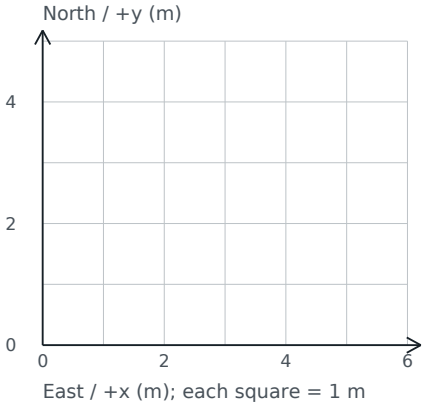
A runner completes one **400 m** lap of a circular track, then runs another **100 m**.
1. **Sketch the full trip** and **mark the final point**.
2. Is the displacement magnitude **500 m**?
3. **Explain how the track geometry** determines the start-to-finish distance.




**Marzano check - circle:** ① ② ③

### 06 Different paths, same endpoints

A robot must travel from (**0**, **0**) m to (**4**, **3**) m.
1. **Draw two routes** on the grid: one using only *east*/*north* legs and one direct route.
2. **Find each path length** and the displacement magnitude both routes share.




**Marzano check - circle:** ① ② ③

### 07 Zero and nonzero

Can displacement be zero while distance is nonzero? Can distance be zero while displacement is nonzero? **Explain why each answer** is correct.

**Marzano check - circle:** ① ② ③

### 08 Review: connect an earlier idea

A fictional observing report records **3** images of a moving object but gives no distance or size calibration. **State a supported observation** and **explain why a precise impact** prediction is unsupported.
Separate what was observed from what can be inferred.

| Observation | Supported claim | Missing information |
|---|---|---|
| ___ | ___ | ___ |
| ___ | ___ | ___ |

**Marzano check - circle:** ① ② ③

### 09 Exit check

Start at **+5 m**, walk to **-2 m**, and finish at **+1 m**.
1. **Calculate distance** and displacement.
2. **Explain why neither equals final** position.

**Marzano check - circle:** ① ② ③

### 10 STRETCH: test the prediction

In one dimension, start at **-2 m** and end at **+3 m** with total distance **11 m**. **Find two possible single-turn positions** and **draw both routes**.




**Marzano check - circle:** ① ② ③